

TECHNOLOGY

SHAPING A BETTER FUTURE

How digital tools change the way students learn, teachers teach and communities share information. A mixed-methods study with 420 survey respondents and 24 interviews in three Indonesian cities.



ABSTRACT

This study examines how technology adoption affects learning outcomes, teaching practice and civic participation. We combine a survey of 420 respondents with 24 interviews and a document analysis of school policies. Adoption is high but uneven: access is no longer the main barrier, skills and support are.

INTRODUCTION

Since 2020 every school in the sample has an internet connection, yet only 41% of teachers use digital tools weekly. This gap between access and use is the starting point of the study.

RESEARCH QUESTIONS

- 1 How does device use relate to learning outcomes?
- 2 Which factors explain teacher adoption?
- 3 What support do schools need most?
- 4 How do communities use school technology?

BENEFITS OF TECHNOLOGY

- ✓ Higher reading scores **+12%** with weekly device use
- ✓ Faster adoption **2x** in online teacher communities
- ⌚ Time saved **3 h** per teacher per week on admin
- ✓ Parents reached **68%** through school messaging apps

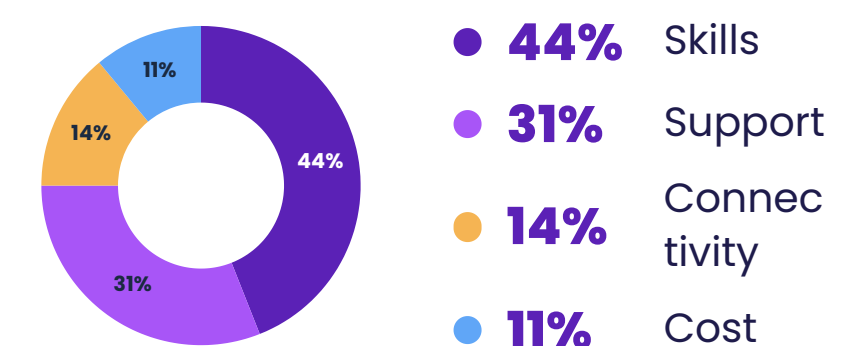
METHODOLOGY

- Surveys**
420 respondents in 3 cities, 32 questions
- Interviews**
24 teachers, students and officials
- Data analysis**
Regression and thematic coding

KEY FINDINGS

- 1. Research**
Students with weekly device use score 12% higher in reading tests
- 2. Communication**
Teachers who join online communities adopt new tools twice as fast
- 3. Challenges**
Skills and technical support, not connectivity, limit adoption

CONCLUSION



Investment should move from hardware to training and support.

“ Skills leads with 44% of the total. ”

RECOMMENDATIONS

- For students**
Weekly guided practice with digital tools, not only free access
- For educators**
Peer communities and paid time for training
- For society**
Open school labs after hours for parents and neighbours
- For government**
Fund technical support staff, one per five schools



TECHNOLOGY TODAY, A BETTER TOMORROW.

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